

Prime-Wave Phase Transition and Conditional Proof Interfaces: A Second Interface-Led Approach to the Riemann Hypothesis

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Author: Takuya Sogawa

ORCID: 0009-0009-7499-2595

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1. Abstract

This technical note proposes a second Interface-Led Architecture (ILA) approach to the Riemann Hypothesis. It does not claim to prove the hypothesis. Instead, it defines a prime-wave phase-transition model and a set of conditional proof interfaces that make explicit which analytic bridges, regularization steps, and non-circular estimates would be required before the heuristic could be upgraded into a rigorous theorem.

The model interprets prime-side oscillations as finite prime-wave modes and treats the critical line $\sigma = 1/2$ as a candidate phase boundary between polynomially divergent prime-mode energy and finite-energy damping. Unlike the prior ILA-RH note, which presented a system-governance blueprint, this paper focuses on proof boundaries: finite prime sums, Perron-type interfaces, the Chebyshev function, Riemann's explicit formula, symmetry constraints from the completed zeta function, and the failure modes that prevent the current model from proving RH.

The main contribution is not a proof of the Riemann Hypothesis, but a role-and-contract decomposition of the proof obligations. In ILA terms, prime modes are treated as Providers, explicit-formula observations as Observers, regularization and contour transformations as Operators, and non-circular analytic requirements as Contracts. This paper thereby provides a structured research program for evaluating whether prime-side criticality, explicit-formula consistency, and functional-equation symmetry can be connected without assuming RH-equivalent error bounds. In addition, it introduces a minimal IUniformPhaseProvider specification as a first separable contract for the prime-phase source.

2. Position and Non-Proof Boundary

Note. The arguments presented here are heuristic and conceptual; they do not constitute rigorous proofs. Rigorous justification requires establishing random-phase estimates, non-circular error bounds, explicit-formula bridges, and operator-theoretic formulations.

The Riemann Hypothesis states that the non-trivial zeros of the Riemann zeta function lie on the critical line $\text{Re}(s) = 1/2$. This remains an open problem. The purpose of the present paper is therefore deliberately limited: it introduces a proof-interface architecture rather than a proof.

This paper builds on the earlier technical note *Interface-Led Architecture and the Riemann Hypothesis: A System-Governance Heuristic for Prime Distribution and Dynamic Equilibrium*. The earlier paper proposed a system-governance interpretation of the relation between prime-side oscillations and zero-side cancellation. The present paper is not a new version of that note. It is a distinct second approach focused on prime-wave phase transition, explicit-formula bridging, and conditional proof interfaces.

The guiding principle is:

$$\text{Heuristic model} \neq \text{proof of RH.}$$

A valid proof would need to replace each heuristic bridge below by a precise analytic theorem.

3. ILA Interpretation

Note. In this paper, ILA is used as a design language for decomposing proof obligations. It is not a substitute for analytic number theory.

The ILA interpretation used here is intentionally minimal.

ILA role	Number-theoretic interpretation	Function in the model
Provider	Prime modes, von Mangoldt weights, finite Dirichlet signals	Supply oscillatory arithmetic data
Observer	Chebyshev functions, explicit formula, zero sums	Observe macroscopic spectral constraints
Operator	Perron integrals, Abel/Mellin regularization, contour shifts	Transform finite signals into analytic interfaces
Contract	Non-circular error bounds, exchangeability, symmetry consistency	Define what must be proved
Fail-Closed	Unjustified exchange, RH-equivalent estimate, unconstructed operator	Stop the proof claim

This framing does not prove anything by itself. Its value is organizational: it separates what is supplied, what is observed, what is transformed, and what remains to be proved.

4. Prime-Wave Finite Model

Note. The finite prime-wave model below is an observation model, not an identity for $\zeta(s)$ inside the critical strip.

Let p_n denote the n -th prime and let

$$\theta_n(t) = -t \log p_n.$$

A finite prime-wave signal may be written as

$$D_N(s) = \sum_{n=1}^N p_n^{-s} = \sum_{n=1}^N p_n^{-\sigma} e^{-it \log p_n}.$$

This finite signal is useful for visualizing prime-side oscillation, but it should not be identified with $\zeta(s)$ in the critical strip. The Euler product and the ordinary Dirichlet series representation of $\zeta(s)$ are initially valid in their convergence domains; values in the critical strip are obtained by analytic continuation.

For phase-boundary analysis, a safer energy proxy is

$$E_N(\sigma) = \sum_{n=1}^N p_n^{-2\sigma}.$$

As $N \rightarrow \infty$, the formal behavior is:

- $\sigma < 1/2$: the prime-mode energy diverges strongly;
- $\sigma = 1/2$: the energy is at a critical logarithmic boundary;
- $\sigma > 1/2$: the energy is damped into a finite-energy phase.

Thus $\sigma = 1/2$ is not presented here as a proved zero locus. It is presented as a critical boundary in a finite-energy heuristic.

4.1 IUniformPhaseProvider: Minimal Contract for an Independent Prime-Phase Provider

Note. This subsection does not complete a proof. It defines a first separable contract on the path toward a proof: the uniformity requirement for the prime-phase source. The purpose is to isolate the behavior of the Provider from analytic continuation, explicit-formula exchangeability, zero correlations, and RH-equivalent error estimates.

We define the arithmetic frequency source as an abstract interface:

```
interface IUniformPhaseProvider
```

The Provider accepts the following inputs:

- $N \in \mathbb{N}$: the number of finite prime modes to be evaluated;
- $t \in \mathbb{R}$: the temporal phase parameter;
- p_n : the n -th prime.

Each prime mode supplies the phase

$$\theta_n(t) = -t \log p_n \pmod{2\pi}.$$

Contract 1: Asymptotic equidistribution on a fixed-dimensional torus For any fixed N , the frequency vector

$$(\log p_1, \dots, \log p_N)$$

is linearly independent over $\mathbb{Q}$. Indeed, if

$$\sum_{n=1}^N q_n \log p_n = 0, \quad q_n \in \mathbb{Q},$$

then clearing denominators gives

$$\prod_{n=1}^N p_n^{m_n} = 1, \quad m_n \in \mathbb{Z}.$$

By unique factorization, every m_n must be zero, hence every q_n is zero.

By a Kronecker-Weyl type equidistribution theorem, this rational independence implies that, for every continuous function

$$g : \mathbb{T}^N \rightarrow \mathbb{R},$$

the temporal average is expected to converge to the Haar-space average:

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T g(\theta_1(t), \dots, \theta_N(t)) dt = \int_{\mathbb{T}^N} g(x) dx.$$

This is the minimal independence contract required of the prime-phase Provider.

Contract 2: Minimal mean-square cancellation bound From the equidistribution contract, for fixed N , the time-averaged cross terms vanish. Therefore, for the critically weighted finite sum, one formally obtains

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \left| \sum_{n=1}^N p_n^{-1/2} e^{i\theta_n(t)} \right|^2 dt = \sum_{n=1}^N \frac{1}{p_n}.$$

By a Mertens-type estimate, the right-hand side grows on the order of

$$\sum_{p \leq p_N} \frac{1}{p} \sim \log \log p_N.$$

Thus a weak Provider-side bound can be stated as

$$\frac{1}{T} \int_0^T \left| \sum_{n=1}^N p_n^{-1/2} e^{i\theta_n(t)} \right|^2 dt \leq C \log \log p_N \quad (T \rightarrow \infty).$$

However, this contract is a time-average statement for fixed N . It does not automatically provide an effective estimate uniform in N , a finite-time discrepancy bound in T , or exclusion of exceptional values of t . Those remain proof obligations.

Dependency isolation The `IUniformPhaseProvider` contract is defined independently of:

1. analytic continuation of $\zeta(s)$;
2. RH-equivalent error bounds for prime-counting or Chebyshev functions;
3. zero correlations, Montgomery pair correlation, or random matrix theory;
4. exchangeability of zero sums in the explicit formula.

Accordingly, this contract extracts the minimal phase-source condition supplied by the prime-side Provider without assuming the Riemann Hypothesis.

Mock-provider testability The specification can also be tested with a Mock Provider: for example, an explicitly chosen rationally independent artificial frequency vector, or a provider that directly samples independent uniform phases.

The purpose of a Mock Provider is not to prove RH. Its purpose is to test, independently of the analytic machinery, whether the uniform-phase contract alone reproduces toroidal probability trajectories and mean-square cancellation under the critical weighting $p_n^{-1/2}$.

This separation yields the following proof architecture:

uniform phase contract
 → prime-mode criticality
 → explicit-formula bridge
 → conditional proof interface.

This subsection addresses only the first arrow.

5. Phase Transition Interpretation

The model suggests a phase diagram:

Region	Energy behavior	Heuristic interpretation	Proof status
$\sigma < 1/2$	divergent prime-mode energy	under-damped / runaway Provider phase	heuristic
$\sigma = 1/2$	critical logarithmic boundary	scale-critical governance boundary	heuristic
$\sigma > 1/2$	finite-energy damping	over-damped / bounded phase	heuristic

This phase diagram does not exclude zeros away from the critical line. Macroscopic energy control is not the same as local zero exclusion. A rigorous proof would require showing that any zero with $\beta \neq 1/2$ violates explicit-formula consistency, functional-equation symmetry, or a non-circular analytic bound.

6. Proof Boundary Justification: Why the Present Model Remains Heuristic

This section explains why the present model cannot claim to prove the Riemann Hypothesis.

6.1 Finite prime sums are not identical to $\zeta(s)$ in the critical strip

The Dirichlet series

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$$

and the Euler product

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1}$$

are not identities that can be naively used throughout the critical strip. Values of $\zeta(s)$ in $0 < \sigma < 1$ are obtained by analytic continuation.

Therefore, a finite prime signal such as

$$D_N(s) = \sum_{n=1}^N p_n^{-s}$$

cannot be treated as $\zeta(s)$ itself. It is an observation model for prime-side oscillations, not a replacement for analytic continuation.

6.2 Random-walk intuition does not imply RH

A random-walk model may suggest that a typical vector sum has size comparable to $O(N^{1/2})$. However, RH is not a typical or probabilistic statement. It is a deterministic assertion about all non-trivial zeros.

Probabilistic intuition does not exclude exceptional values of t , exceptional phase alignment, or local cancellation. A proof would require uniform analytic estimates and explicit control of exceptional sets.

6.3 Macroscopic energy control does not exclude local zeros

The phase model describes global or macroscopic energy behavior. However, a local zero is a pointwise cancellation event. Even if an energy average is small, this does not by itself prove that no local zero can occur. Conversely, even if an energy proxy grows, local cancellation may still occur.

Thus prime-mode energy can suggest a critical boundary, but it does not establish zero exclusion.

6.4 A Hilbert-Polya operator has not been constructed

The model is compatible with a Hilbert-Polya style intuition, but it does not construct the required operator. Such a program would require:

1. a Hilbert space $\mathcal{H}$;
2. a self-adjoint operator H on $\mathcal{H}$;
3. a proof that the spectrum of H matches the ordinates γ of zeta zeros;
4. a derivation that self-adjointness forces $\beta = 1/2$.

None of these are completed here.

6.5 Explicit-formula exchangeability is not yet proved

Connecting finite prime sums to zero sums requires exchanging limits, sums, contour integrals, and regularization procedures. This requires uniform convergence, principal value control, Abel/Perron/Mellin regularization, or Tauberian-type conditions.

The present model lists these exchangeability requirements, but does not prove them.

7. Summary of Proof Obligations

Interface	Missing obligation	Required justification
Finite prime signal	relation between $D_N(s)$, $A_x(s)$, and $\zeta(s)$	regularization and explicit-formula bridge

Interface	Missing obligation	Required justification
IUniformPhaseProvider	gap between fixed- N equidistribution and estimates uniform in N, T	discrepancy bounds, finite-time estimates, exceptional- t control
Random phase	average intuition vs deterministic claim	uniform estimates and exceptional-set control
Phase transition	energy boundary vs local zero exclusion	analytic inequalities strong enough to exclude zeros
Operator theory	Hilbert-Polya operator not constructed	Hilbert space, self-adjoint operator, spectral identification
Explicit formula	exchange of limits, sums, and integrals	Abel/Perron/Mellin regularization and convergence proofs

The contribution of the model is to expose these obligations, not to hide them.

8. Explicit Formula Bridge

Note. This section gives a bridge specification, not a completed proof. Its purpose is to identify the analytic transformations that must be justified before finite prime-wave models can be compared with zero-side spectral sums.

To connect microscopic finite prime oscillators to the macroscopic spectrum of zeta zeros, the bridge should pass through the von Mangoldt function, Perron's formula, and Riemann's explicit formula.

Define the finite von-Mangoldt weighted Dirichlet polynomial

$$A_x(s) = \sum_{n \leq x} \Lambda(n) n^{-s}.$$

For $\text{Re}(s) > 1$, the logarithmic derivative of the zeta function satisfies

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \Lambda(n) n^{-s}.$$

Thus $A_x(s)$ is a finite approximation to this logarithmic derivative in its initial convergence domain. It is not the logarithmic derivative itself inside the critical strip.

The Chebyshev function is

$$\psi(x) = \sum_{n \leq x} \Lambda(n).$$

Formally, Perron's formula gives a contour integral representation

$$\psi(x) = \frac{1}{2\pi i} \int_{\sigma_0 - i\infty}^{\sigma_0 + i\infty} \left(-\frac{\zeta'(s)}{\zeta(s)} \right) \frac{x^s}{s} ds, \quad \sigma_0 > 1.$$

After shifting the contour and collecting residues of $-\zeta'(s)/\zeta(s)$, one obtains a standard form of Riemann's explicit formula:

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log(2\pi) - \frac{1}{2} \log(1 - x^{-2}),$$

with the usual caveats concerning discontinuities, truncation, and convergence of the zero sum.

8.1 Bridge Requirement 1: Regularized finite signal

The finite signal $A_x(s)$ must be related to $-\zeta'(s)/\zeta(s)$ through a regularized limiting procedure. The bridge requires:

1. truncation control for $A_x(s)$;
2. a justified Perron integral;
3. contour shifting with controlled horizontal integrals;
4. residue accounting for the pole at $s = 1$, non-trivial zeros, trivial zeros, and constants;
5. a non-circular bound for the residual error $\mathcal{E}(x)$.

8.2 Bridge Requirement 2: Prime-mode energy and zero sums

For prime-only modes, the basic energy proxy is

$$E_x(\sigma) = \sum_{p \leq x} p^{-2\sigma}.$$

For the explicit formula, a von-Mangoldt weighted version is more natural:

$$E_x^{\Lambda}(\sigma) = \sum_{n \leq x} \Lambda(n)^2 n^{-2\sigma}.$$

The bridge must show how the scale behavior of these prime-side energies is reflected in the zero-side sum

$$\sum_{\rho} \frac{x^{\rho}}{\rho}.$$

This correspondence is not automatic. It requires a precise regularization and comparison theorem.

8.3 Bridge Requirement 3: Zero-mode scale consistency

A zero $\rho = \beta + i\gamma$ contributes a mode

$$\frac{x^{\rho}}{\rho} = \frac{x^{\beta} e^{i\gamma \log x}}{\rho}.$$

If $\beta > 1/2$, this mode has amplitude growing faster than $x^{1/2}$. The functional equation pairs such a zero with a reflected zero. The proof interface asks whether this asymmetric amplitude pair can be made inconsistent with prime-side criticality and explicit-formula consistency.

The desired bridge condition is:

$$\text{prime-side critical boundary} \iff \text{zero-spectrum scale consistency}.$$

This condition is a proof obligation, not a theorem established here.

9. Conditional Proof Interface

Note. This section does not prove RH. It defines a conditional interface for avoiding circularity and identifying the exact analytic obligations required by the model.

9.1 Precondition: no RH-equivalent estimate at the start

The proof process must not begin by assuming an error bound equivalent to RH, such as an unqualified $O(x^{1/2+\varepsilon})$ estimate for prime-counting or Chebyshev-type functions.

Instead, it should begin with unconditional information, such as known zero-free regions and weaker prime number theorem error terms. A generic initial form can be expressed as

$$\psi(x) = x + O(x^\theta), \quad 1/2 < \theta < 1.$$

The value $\theta = 1/2$ is not assumed; it is the target boundary.

9.2 Conditional invariant

The conditional proof interface may be written as follows:

Step 1 (Precondition): $\psi(x) = x + O(x^\theta), \quad 1/2 < \theta < 1,$

Step 2 (Prime-mode feedback): $E_x^\Lambda(\sigma) = \sum_{n \leq x} \Lambda(n)^2 n^{-2\sigma},$

Step 3 (Bridge consistency): $E_x^\Lambda(\sigma)$ is compared with $\sum_{\rho} x^\rho / \rho,$

Step 4 (Contradiction test): $\beta > 1/2$ is tested for scale inconsistency,

Step 5 (Target invariant): $\theta \rightarrow 1/2$ is the proof target.

Step 5 is not a conclusion of this paper. It is the target invariant that a future proof must derive.

9.3 Fail-closed bootstrap

The bootstrap is fail-closed:

```
[unconditional zero-free information]
|
v
[weak error bound  $O(x^\theta)$ ,  $1/2 < \theta < 1$ ]
|
v
[von-Mangoldt weighted prime-mode energy]
|
v
[explicit-formula scale consistency]
|
v
[test  $\beta > 1/2$  for asymmetric amplitude leakage]
|
v
[prove contradiction with analytic continuation,
functional-equation symmetry, and explicit-formula consistency]
|
```

v

[target: theta = 1/2 type boundary]

If any required estimate, exchangeability condition, or regularization step fails, the argument stops and no proof claim is made.

9.4 Symmetry-breaking contradiction interface

Assume for contradiction that there is a non-trivial zero

$$\rho_0 = \sigma_0 + i\gamma_0, \quad \sigma_0 > 1/2.$$

The functional equation for the completed zeta function implies a reflected zero

$$\rho_1 = 1 - \overline{\rho_0}.$$

The corresponding explicit-formula modes have amplitude scales

$$\frac{x^{\sigma_0}}{|\rho_0|} \quad \text{and} \quad \frac{x^{1-\sigma_0}}{|\rho_1|}.$$

The desired contradiction would show that the asymmetric amplitude growth caused by $\sigma_0 > 1/2$ cannot be reconciled with the prime-side critical boundary and the explicit-formula bridge.

Formally, the target implication is

$$\beta > 1/2 \Rightarrow \text{scale inconsistency with prime-mode criticality} \Rightarrow \text{contradiction with explicit-formula consistency}.$$

This implication is not proved in this paper. It is the central conditional proof interface.

10. Open Problems

A future rigorous development must solve at least the following problems:

1. prove non-circular random-phase or cancellation estimates for prime modes;
2. connect finite prime-wave energy to $\psi(x)$ through Perron or Mellin interfaces;
3. control all truncation and contour-shift error terms without RH-equivalent assumptions;
4. establish whether off-critical zeros create unavoidable scale inconsistency;
5. account for possible cancellations among multiple zero modes;
6. construct or reject a Hilbert-Polya type operator corresponding to the prime-wave model;
7. produce reproducible computational experiments that test the bridge requirements without overstating their meaning.

Implementation references, GitHub links, commit hashes, and function-to-interface mapping tables are intentionally deferred to a future version.

11. Conclusion

This paper introduced a second ILA approach to the Riemann Hypothesis. It reframes the problem not as a completed proof but as a set of conditional proof interfaces connecting prime-wave phase transition, explicit-formula bridging, and functional-equation symmetry.

The central insight is that $\sigma = 1/2$ can be interpreted as a prime-mode critical boundary, but this interpretation alone does not prove that all non-trivial zeros lie there. A proof would require showing that any off-critical zero breaks scale consistency in a way that cannot be repaired by cancellation, regularization, or contour deformation.

The contribution of this note is therefore architectural and analytic: it decomposes a possible route to RH into explicit Providers, Observers, Operators, Contracts, and Fail-Closed proof boundaries. It makes clear what remains unsolved and where a genuine proof would have to enter.

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